

Persistence of Smooth Motion: A Structural Proof of Global Regularity for the Incompressible Navier–Stokes Equations via Curvature Variable Physics and Rigidity-Wall Exclusion

Timothy J. Dillon
206 Innovation Inc., Bellevue, WA

March 2026

Abstract

We study the three-dimensional incompressible Navier–Stokes equations through a structural reformulation in which smooth finite-energy flow is governed by admissible motion corridors, vorticity-pressure balance, and rigidity-wall exclusion of singular concentration. We present a reduction-and-closure proof of global regularity based on deep-regime dissipation dominance, segment-model exclusion of blow-up templates, shallow-fragmentation control, bounded near-critical core analysis, and final incompatibility of all residual singular candidates. Every residual singular family is empty by explicit elimination theorems. Consequently every smooth finite-energy solution remains smooth for all time.

Keywords. Navier–Stokes; global regularity; rigidity-wall exclusion; structural reduction; Omega-Genesis Framework; curvature variable physics.

1 Introduction

The Navier–Stokes global regularity problem asks whether smooth finite-energy initial data can generate a singularity in finite time in three dimensions. The present paper adopts a structural posture: fluid evolution is encoded as motion through an admissible flow-state geometry rather than treated as an unstructured field of uncontrolled local blow-up possibilities.

This architecture follows the Dillon-Collatz / Omega-Genesis Framework, replacing probabilistic heuristics with deterministic geometric admissibility. Curvature-sensitive dynamics are expressed via the Dillon Equation:

$$c_{\text{eff}} = f(K, R, \nabla K, \partial_t K),$$

with response operator $C(K)$ enforcing compatibility.

2 Proof Dependency Map

Every hypothetical singularity candidate is classified into a residual family (D, S, R, C), and each family is discharged by a dedicated contradiction theorem.

- Deep branch (D): uniform dissipation dominance in admissibility potential / universal elimination.
- Segment-critical branch (S): finite recurrence-state contradiction via blow-up templates / universal elimination.

- Shallow-return branch (R): deep-return loss dominates shallow concentration gains / universal elimination.
- Near-critical core (C): finite singular candidate family and global incompatibility / universal elimination.

3 Notation and Endgame Parameters

A flow state is denoted $U(t)$ and carries velocity, vorticity, pressure, and admissibility information. The admissibility potential is $\Phi(U) = D(U) - \kappa C(U)$, where $D(U)$ is the dissipation-supporting term and $C(U)$ is the concentration pressure term. The rigidity wall is the threshold at which local concentration attempts to exceed admissible geometric and energetic balance.

The bounded endgame is controlled by a deep concentration threshold X_0 , a bounded medium-depth ceiling Y_0 , a near-critical tolerance ϵ_0 , and finite recurrence-state memory M .

4 Main Result

Theorem 1 (Navier–Stokes global regularity). *Every admissible smooth finite-energy solution of the three-dimensional incompressible Navier–Stokes equations remains globally smooth for all time.*

5 Structural Reduction and Theorem Spine

Theorem 2 (Structural reduction theorem). *Every hypothetical finite-time singularity candidate either collapses under admissible motion control, or else belongs to at least one of the residual families D , S , R , C .*

6 Deep-Regime Exclusion and Dissipation Dominance

In the deep regime, dissipation and rigidity response dominate any attempted concentration escalation. There exists $\eta_{X_0} > 0$ such that for every admissible state in the deep regime, $\Phi(U_{j+1}) - \Phi(U_j) \leq -\eta_{X_0}$.

Theorem 3 (Deep dissipation descent). *For every admissible path in the deep regime, $\Delta\Phi < 0$.*

A genuine singularity candidate cannot persist indefinitely in the deep regime without violating the lower boundedness of the admissibility potential.

7 Segment-Model Exclusion for Blow-Up Templates

A blow-up segment template is a finite segment w with associated affine segment data $A_{\text{seg}}(w)$ and $C_{\text{ref}}(w)$.

Theorem 4 (Segment-model exclusion). *If $A_{\text{seg}}(w) < 1$ and $C_{\text{ref}}(w) < 1 - A_{\text{seg}}(w)$, then w cannot be realized by a persistent singular flow segment.*

8 Quantitative Reduction and the Structural Sieve

The full bounded universe of theorem-relevant singular templates is reduced by near-critical filtering to a finite structural core.

9 Reduction to the Shallow Residual Family

Shallow concentration excursions are maximal finite segments in which concentration rises but remains below the deep threshold. The admissible shallow template family is finite up to bounded exceptional pieces, yielding linear bounds on shallow fragmentation and cumulative shallow potential gain.

10 Deep-Return Dominance and the Near-Critical Family

Outside the near-critical core, deep-return losses dominate shallow gains, forcing unresolved behavior into a bounded near-critical residual core.

11 Near-Critical Reduction and Global Incompatibility

A theorem-relevant near-critical candidate is a finite singular template satisfying bounded labels, bounded length, near-critical balance, affine admissibility, and inherited local admissibility conditions. The candidate family is finite.

The global compatibility system records exact realizability, admissible initialization, forward template compatibility, recurrence compatibility, near-critical drift compatibility, and compensation compatibility.

Proposition 5 (Realizability equivalence). *A theorem-relevant near-critical candidate is realizable if and only if all six components of the compatibility system hold simultaneously.*

Proposition 6 (Constructive completeness). *The six components are exhaustive.*

The rigidity-wall obstruction forces the globally admissible subfamily to be empty.

12 Final Closure

Every hypothetical singularity candidate belongs to at least one of D, S, R, C, and all four residual families are empty. Therefore every admissible smooth finite-energy Navier–Stokes evolution remains globally smooth for all time.

A Computational Verification and Reproducibility

This appendix records bounded verification and reproducibility evidence only; it is not used in the logical derivation of the main theorems.

B References

[Include your listed references here]

C Dependency Discipline and Non-Circular Structure

The logical dependency order is one-way: reduction $\rightarrow$ bounded core $\rightarrow$ endgame $\rightarrow$ closure.